

Core Banking & STP Underwriting Migration

End-to-end project planning and control using MS Project, combining Hybrid Agile–Waterfall delivery with schedule, resource, cost, risk and cutover management.

180 Days	baseline duration
■3.80 Cr	approved budget
1.4M	records to migrate
≤48 hrs	production cutover
Day 75	status control point

THE PROJECT

Plan a controlled migration from a legacy Core Banking System (CBS) to a cloud-native CBS with automated Straight-Through Processing (STP) underwriting, while protecting data integrity, regulatory compliance and business continuity.

PROJECT OBJECTIVES

- Reduce underwriting turnaround from up to 48 hours to a target of <5 minutes through automated STP decisioning.
- Reduce application abandonment from 32% to 8% through faster and more consistent credit decisions.
- Migrate 1.4M legacy records with 100% reconciliation / zero variance as the acceptance target.
- Complete production cutover within a maximum 48-hour controlled window with defined recovery controls.

PROJECT MANAGEMENT APPROACH

Area	How it was handled
Scope & governance	Project Charter, objectives, constraints, acceptance criteria and RACI accountability.
Planning	WBS decomposed the project into work packages and activities; durations and predecessors were entered into MS Project.
Scheduling	Gantt chart tracked planned vs actual progress; AON/CPM identified the controlling sequence and float.
Delivery model	Hybrid Agile–Waterfall: governance, compliance, architecture, migration, integrated testing and closure followed Waterfall controls; STP engine was delivered through four two-week Agile sprints.
Control	Resource leveling, crashing and Earned Value Analysis (EVA) were used to manage resource, time and cost trade-offs.
Execution & closure	Risk register, 48-hour cutover control room, Go/No-Go checkpoint, reconciliation and formal project closure.

SCOPE ARCHITECTURE

The Work Breakdown Structure (WBS) organized the project into seven major work packages: 1.1 Initiation & Regulatory Compliance; 2.0 Architecture & Cloud; 3.0 STP Underwriting Engine; 4.0 Database Migration & ETL; 5.0 Testing & Security; 6.0 48-Hour Live Cutover; and 7.0 Stabilization & Closure.

PROJECT LENS

The project demonstrates how MS Project converts business requirements into a controllable schedule and then connects scope, dependencies, resources, cost, performance, risk and closure into one management system.

Turning scope into a controllable schedule

The project used WBS, Gantt and Activity-on-Node / Critical Path Method analysis to understand sequencing, dependencies, float and schedule risk.

WORK BREAKDOWN STRUCTURE (WBS)

WBS means Work Breakdown Structure. It decomposes the project from the overall objective into manageable work packages and activity groups. The project used Level 1 as the project, Level 2 as work packages and Level 3 as deliverable/activity groups.

WBS	Work package	Illustrative activities
1.1	Initiation & Regulatory Compliance	Project Charter; RBI Compliance; Audit Review
2.0	Architecture & Cloud	Cloud VNet; Security Hardening; Target DB Schema
3.0	STP Underwriting Engine	Sprint 1 Bureau Connectors; Sprint 2 Credit Policy; Sprint 3 APIs; Sprint 4 Dashboard
4.0	Database Migration & ETL	Data Cleansing; ETL Pipeline; Delta Synchronization
5.0	Testing & Security	UAT; Security Audit; Migration Dry Runs
6.0	48-Hour Live Cutover	Freeze; Final ETL; Reconciliation; Go/No-Go; Traffic Routing; Go-Live
7.0	Stabilization & Closure	Handover; Financial Closure; Lessons Learned; PMO Archival

GANTT — PLANNED VS ACTUAL CONTROL

The Gantt chart compares the baseline plan with actual progress at the Day-75 status date. Completed activities included Charter/Steering setup, RBI Compliance & Audit Review, Cloud VNet & Security Hardening, Target DB Schema & Sharding Configuration and Legacy Data Cleansing. Sprint 1 was marked Lagging, Sprint 3 / ETL was In Progress, while later testing, cutover and closure activities remained Planned.

CONTROL SIGNAL

SPI = 0.94 at Day 75, indicating the project was progressing below the planned schedule pace. The Gantt therefore highlights where management attention was required.

AON / CPM — IDENTIFYING THE CONTROLLING PATH

AON means Activity-on-Node. CPM means Critical Path Method. Each node represents an activity and its dependency links. Forward pass calculates Early Start (ES) and Early Finish (EF); backward pass calculates Late Start (LS) and Late Finish (LF). Slack / Float shows how much an activity can move without moving the project finish.

Term	Meaning
ES	Earliest Start — earliest the activity can begin.
EF	Earliest Finish — ES + activity duration.
LS	Latest Start — latest start without delaying project finish.
LF	Latest Finish — latest finish without delaying project finish.
Slack / Float	Scheduling flexibility; LS – ES.
Critical Path	Sequence with zero slack; delay to a critical activity can delay project completion.

PROJECT NETWORK FINDING

The real-scenario AON view shows a 180-day project length in the MS Project summary and identifies a zero-slack critical sequence running through the major delivery chain. The project also highlights that some migration activities retain limited scheduling flexibility, while activities such as 4.1 Legacy Data Cleansing and 4.2 ETL Pipeline & Delta Sync were shown with scheduling float.

MANAGEMENT INTERPRETATION

Critical-path analysis tells the project manager where delay matters most; float tells where limited schedule flexibility can be used before changing the project finish.

Controlling the project, not just planning it

After creating the schedule, the project applied resource management, time–cost trade-offs, Earned Value Analysis, risk controls and formal closure.

RESOURCE LEVELING

Resource leveling resolves situations where the same specialist is required by overlapping activities beyond their available capacity. In the modeled scenario, the Lead Cloud Architect reached 160% over-allocation during Weeks 8–10. The selected response was to shift activity 3.3 by four business days, using available float and bringing utilization back to ≤100% without changing the critical path.

Resource	Type	Maximum units	Rate
Project Manager	Work	100%	■2,500/hr
Lead Cloud Architect	Work	100%	■3,500/hr
DBA Specialist	Work	200%	■2,800/hr
API Engineers	Work	400%	■2,200/hr
Compliance Consultant	Work	50%	■4,500/hr

CRASHING — TIME VS COST

Crashing means reducing an activity's duration by adding cost/resources when schedule compression is justified. The simulator tested three activities and compared their incremental cost slopes.

Activity	Normal → Crash	Incremental cost	Cost slope
Cloud VNet & Security Hardening	15d → 10d	■6.0L	■1.20L/day
ETL Pipeline & Delta Sync	20d → 14d	■7.2L	■1.20L/day
Integrated UAT & Security	30d → 20d	■15.0L	■1.50L/day

CRASHING LOGIC

The scenarios show 5 days saved at ■6.0L, 11 days saved at ■13.2L, and 21 days saved at ■28.2L when the three interventions are progressively combined. The decision rule is to check cost slope and confirm the activity remains on the controlling path before treating it as genuine schedule compression.

EARNED VALUE ANALYSIS (EVA)

EVA / EVM means Earned Value Analysis / Earned Value Management. It integrates scope, schedule and cost at a common control point. At Day 75, the project had PV (Planned Value) of ■1.62 Cr, EV (Earned Value) of ■1.523 Cr and AC (Actual Cost) of ■1.479 Cr.

Metric	Formula	Day-75 result	Interpretation
CV — Cost Variance	EV – AC	+■4.44L	Favourable / under budget
SV — Schedule Variance	EV – PV	–■9.70L	Behind planned schedule
CPI — Cost Performance Index	EV ÷ AC	1.03	Cost efficient
SPI — Schedule Performance Index	EV ÷ PV	0.94	Behind schedule

RISK & 48-HOUR CUTOVER

The risk register prioritized data corruption during cutover, DBA attrition, Bureau API latency, RBI non-compliance and branch adoption friction. Responses included checksum validation and hourly rollback snapshots, a secondary certified DBA partner, CIBIL-to-Experian failover, RBI weekly audit plus legal retainer, and a two-week sandbox/certification approach.

Cutover stage	Control
T+0–2	Freeze + snapshot
T+2–12	Final delta ETL
T+12–20	Reconciliation
T+20–28	Synthetic loan + Go/No-Go decision
T+28–38	Routing cutover
T+38–48	Pilot + Go-Live

CLOSURE & FINAL OUTCOME

- Administrative acceptance: Steering Committee and Head of Credit sign-off, 14-day hypercare, zero Sev-1 defects and operational handover.
- Contractual & financial close: vendor milestone settlement, cloud consumption reconciliation, contractor retention release and final budget reconciliation.
- Lessons learned & PMO closure: baseline-vs-actual archive, post-mortem, Bureau API sandbox learnings and Enterprise PMO repository update.

FINAL PROJECT TAKEAWAY

A complete project-control cycle: define scope → build WBS → schedule dependencies → identify critical path → level resources → evaluate crashing → measure EVA performance → manage risk → execute controlled cutover → close and capture lessons.

KEY PM TERMINOLOGY

CBS = Core Banking System • STP = Straight-Through Processing • WBS = Work Breakdown Structure • AON = Activity-on-Arrow • CPM = Critical Path Method • Gantt = schedule visualization • EVA/EVM = Earned Value Analysis/Management • PV = Planned Value • EV = Earned Value • AC = Actual Cost • CV = Cost Variance • SV = Schedule Variance • CPI = Cost Performance Index • SPI = Schedule Performance Index • ETL = Extract, Transform, Load • UAT = User Acceptance Testing • DBA = Database Administrator • RBI = Reserve Bank of India • PMO = Project Management Office • RPO = Recovery Point Objective • RTO = Recovery Time Objective.